ARHAAN JAIN

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EDUCATION

University of Pennsylvania, Master of Science in Robotics **Aug 2026–May 2028**

- Relevant courses: Machine learning, Machine Perception

National University of Singapore, Bachelor of Engineering in Mechanical Engineering **Aug 2022–Jul 2026**

- GPA: 4.68/5.00 (Highest Distinction)
- Dean's List in AY2025/26 Semester 1 (Top 5% of cohort)
- Minor in Management, Specialization in Robotics
- Relevant courses: Soft Robotics, Robotics System and Design, Deep Learning for Robotics
- Awarded the President's Honor Roll at NUS College, the Honors College of NUS

RESEARCH EXPERIENCE

Research Assistant, Figueroa Robotics Lab **Sep 2026–ongoing**

- Working on a project focusing on making state-of-the-art manipulation policies human-interactive

Research Assistant, Multi-Agent Robotic Motion Lab (MARMot Lab) **Jul 2025–Jul 2026**

- Developed a sim-to-real pipeline using actuator networks for hierarchical RL-based loco-manipulation; trained and deployed policies on the Unitree Go2 and HEBI Daisy Hexapod.
- Developed and deployed a reference-conditioned tracking policy for X-Morph, enabling human-motion transfer to quadruped, hexapod, and quadruped-with-arm morphologies.

Research Intern, Evolution Innovation Lab (Raye Lab) **Jul 2025–Nov 2025**

- Designed and developed a PID-enabled fluidic control board to operate soft robotic actuators with precise pressure regulation (0–150 kPa) and programmable inflation/deflation cycles
- Integrated Arduino GIGA microcontroller with a touchscreen UI for fully user-optimized control of pneumatic channels and automated actuation sequences

Research Fellow, NUS Undergraduate Research **Aug 2024–Dec 2024**

- Carried out research supervised by Professor Tural Khudiyev aimed at creating electronic textiles. *Awarded $5,000 grant for project execution*
- Conducted research on wearable fabric electronics, developing a soldering-based method for fabricating one-dimensional (1D) circuits with 96 micro-LEDs interconnected by conductive wires

MANUSCRIPTS UNDER REVIEW

- Yang, X., Li, P., Li, H., Peng, Y., **Jain, A.**, et al. *HeLoM: Hierarchical Learning for Whole-Body Loco-Manipulation by a Hexapod Robot.* arXiv:2509.23651. Under review, 2026.
- Sharma, R., Sood, S., **Jain, A.**, Kesavamoorthi, S. C., He, C., Sartoretti, G. A. *X-Morph: Human Motion Priors for Scalable Robot Learning Across Morphologies.* arXiv:2606.30290. Under review, 2026.

PROFESSIONAL EXPERIENCE

Roceso Technologies, Engineering Intern, Singapore **Feb 2026–June 2026**

- Conducted R&D on a wearable upper-limb assistive orthosis, using EMG-based muscle-intent detection to control motorized assistance for facilitating arm movement
- Developed work instructions, process control plans and other documentation for in-house fabrication of the flagship product, the EsoGlove Neurorehabilitation system

Pratt & Whitney, Research and Development Intern, Singapore **Jan 2025–May 2025**

- Programmed a robotic arm and machine learning model to automate a manual process, 25% saving on processing time; eliminated ergonomic risks from manual process by innovating on materials used
- Co-developed a machine-learning-powered webapp to digitize engine visual inspections, reducing the manpower hours in by 50%
- Contributed to a factory digital-twin initiative using Leica BLK360 3D scanners

Jyoti CNC Automation Ltd., Application Engineering Intern, India **May 2024–July 2024**

- Optimised cycle time estimation process by automating bottlenecks
- Developed a Flask-based web application to estimate cycle times for CNC setups, facilitates efficient cycle time calculation for a variety of components

ROBOTIC PROJECTS

Pusher Bot, College of Design and Engineering **Jan 2025–Apr 2025**

- Developed a custom MuJoCo–Gym simulation framework for contact-rich robotic pushing
- Implemented and trained a compact actor–critic PPO architecture with GAE under CPU-only constraints
- Performed model evaluation and fine-tuning to raise policy success rate to 92.8% over 1000-episodes

Turtle Soft Robot, College of Design and Engineering **Aug 2024–Dec 2024**

- Designed and built a bio-inspired soft robotic sea turtle using custom-fabricated silicone bellow actuators and an ESP32-based pump–valve control system
- Collaborated in a cross-disciplinary team to deliver a comprehensive technical report, high-quality demo video, and a functional soft robotic prototype demonstrating validated locomotion performance.
- Developed wireless multimodal control, integrating PS4 Bluetooth inputs and Python-based speech-to-text commands via Google's API to execute real-time actuation sequence

Medical Grand Challenge, Yong Loo Lin School of Medicine **March 2024–Aug 2024**

- Designed a functional prototype of our teams' product, PillBuddy: a fully automated pill dispenser to eliminate the need for caretakers in tackling polypharmacy. Used Arduino, 3D printing, and machining to build the prototype
- Finished as finalists in the Nascent Category of Medical Grand Challenge 2024

COMMUNITY SERVICE EXPERIENCES

Overseas Community Service in the Philippines, NUS College **Jan 2023–April 2025**

- Prepared an intensive proposal for a 2-year long service group project in Dauin, Dumaguete as a part of NUS College's capstone Impact Experience programme, received grant of *$S21,000*
- Designed bamboo-based playground and developed a website for fundraising and outreach for community partners, Agape Orphanage
- Conducted field trips to Philippines to learn about bamboo cultivation and construction, actively helped Agape with day-to-day activities
- Working with another orphanage, Bata ng Calabnugan, to help them implement vermicomposting

Deputy Director, Makers' Studio NUS College Club **Nov 2023–Oct 2024**

- Conducted regular checks, maintenance, and repairs, of the facility's equipments
- Hosted and facilitated 5+ training and workshops for students to learn about fabrication processes

ADDITIONAL INFORMATION

- Programming and software: Python, C, SQL, R, MATLAB, SolidWorks, Fusion360, Siemens Plant Simulation
- Robotics and simulation: ROS, MuJoCo, Gazebo, Isaac Lab, Unitree Robotics SDK, HEBI Robotics SDK
- Hardware: Arduino, sensor integration, soldering, 3D printing, CNC Machining, pneumatics, control systems
- Languages: Native in English, Gujarati, and Hindi; professional working proficiency in French; limited working proficiency in German
- Interests: Badminton, football, and squash (competed for inter college games at NUS), Film making (co-created a mini documentary on the Singapore Turf Club)